ROOKIE CONTRACTS & TEAM STABILITY

How rookie-wage-scale spending relates to the volatility of NFL offenses

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EPA/PLAY VOLATILITY · ROOKIE CAP SHARE · DRAFT CAPITAL · TEAM CONTINUITY

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The Question

The 2011 CBA introduced a rookie wage scale, fixing rookie pay to draft position before a player is ever drafted. Cheaper, standardized rookie deals could change how teams behave.

1

Rapid rebuilds

Cheap rookie QBs let teams retool their offense quickly.

2

Faster resets

Teams can afford to reset sooner after a losing stretch.

3

Reallocation

Savings at one position can be spent disproportionately elsewhere.

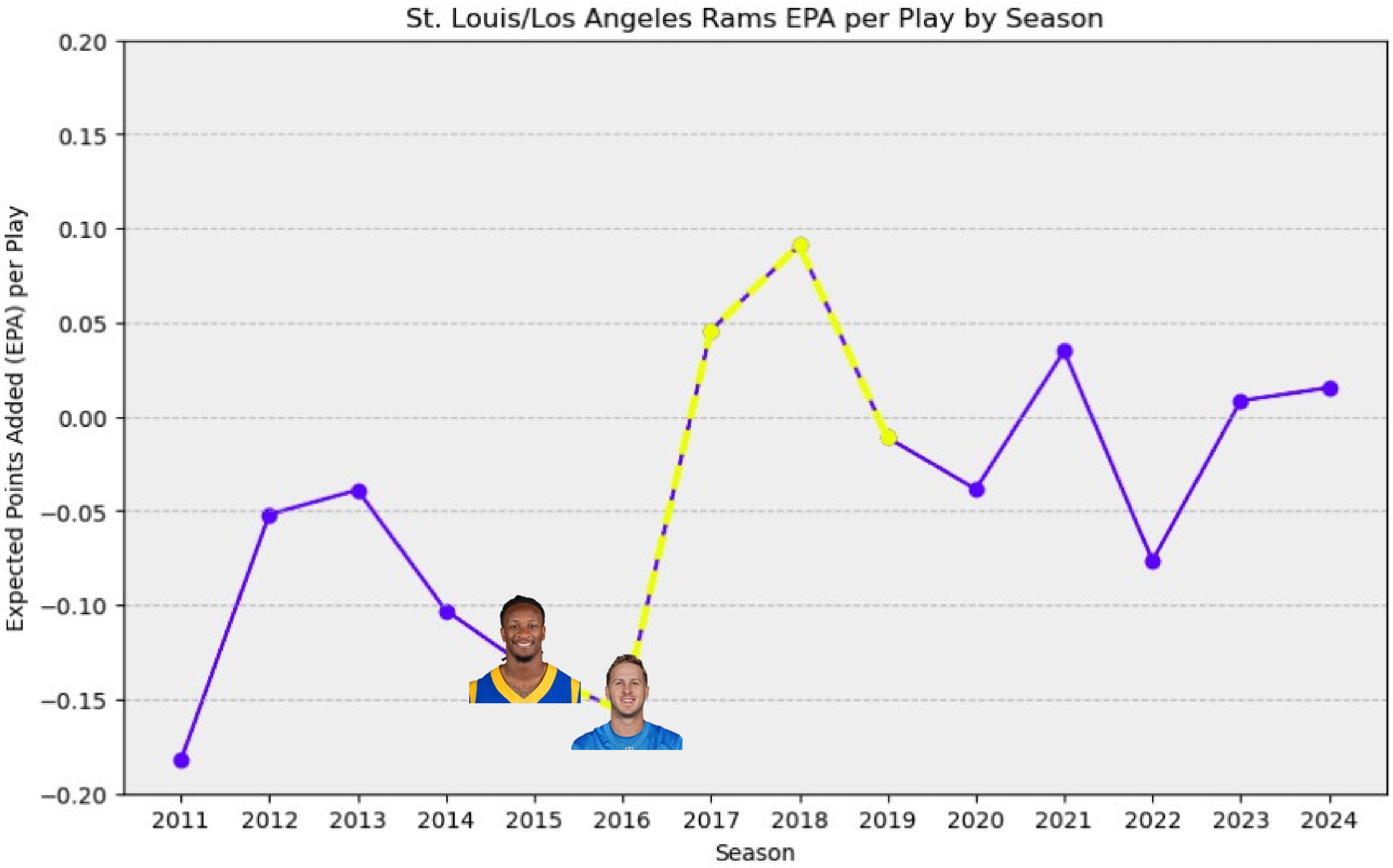
What “volatility” means here

The 4-year forward-looking standard deviation of EPA/play — not whether a team wins, but how consistent its offensive performance is across the length of a rookie's first contract.

Base years: 2011–2015 (rolling windows extend to 2019). 2020 is excluded — COVID introduced offensive noise league-wide that isn't a true reflection of roster construction.

Case Study: Highest Volatility

St. Louis / Los Angeles Rams, 2015–2019



Highest offensive volatility in the sample.

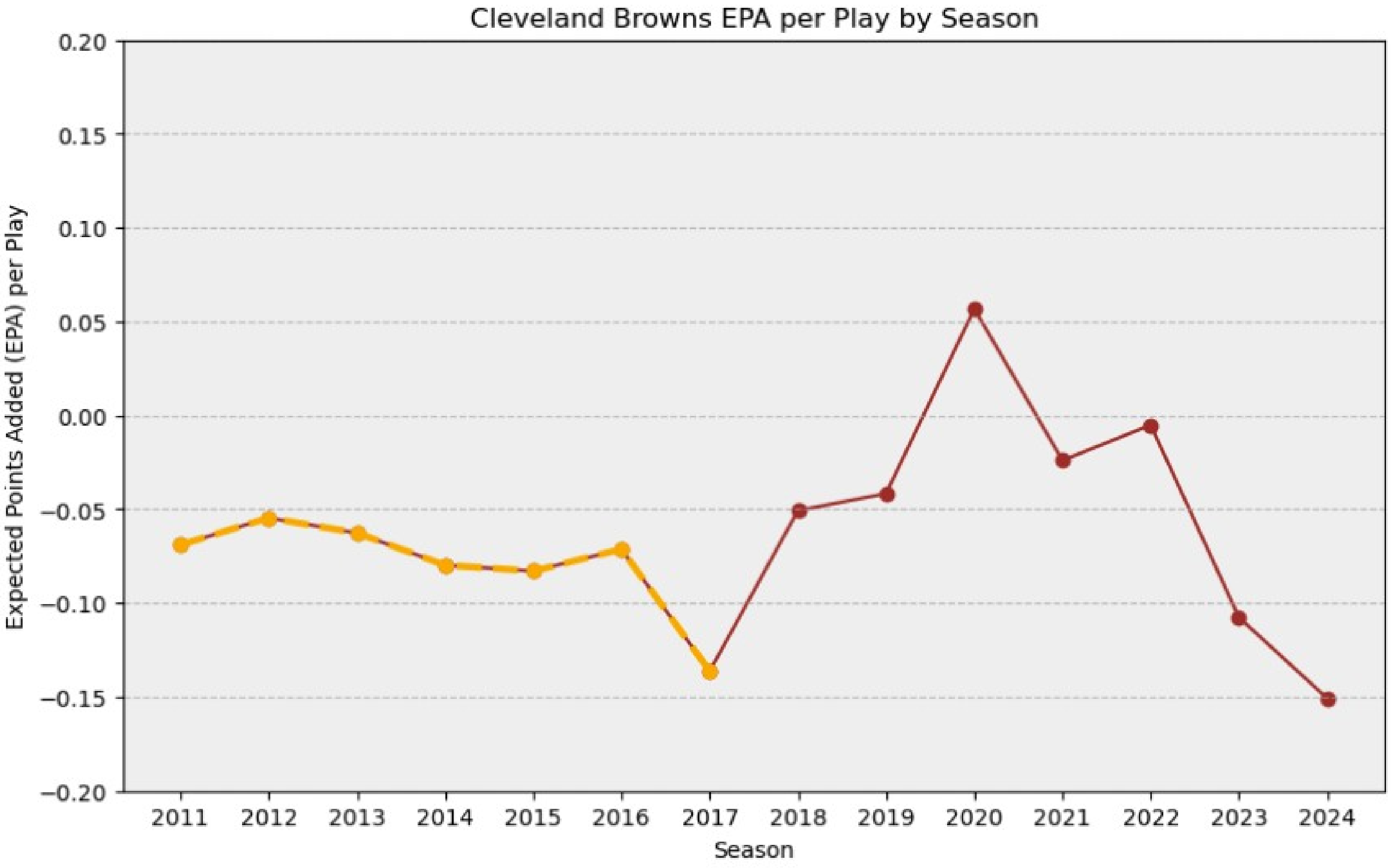
- 2015: draft RB Todd Gurley (OROY, OPOY, 3x Pro Bowl, 3x All-Pro).
- 2016: draft franchise QB Jared Goff (2x Pro Bowl as a Ram).
- Offensive efficiency jumps sharply in the two seasons that follow.

Takeaway

Investment in premium offensive draft picks may play a direct role in how quickly — and how sharply — an offense's performance shifts.

Case Study: Lowest Volatility

Cleveland Browns, overlapping windows 2011–2015 & 2013–2017



Lowest offensive volatility in the sample — but for a different reason.

- Only notable offensive draftee: G Joel Bitonio (2014), later a 2x All-Pro.
- Even a franchise cornerstone couldn't move the needle on a persistently subpar offensive line.
- The Rams pattern doesn't repeat: stability here reflects consistent mediocrity, not a smart bet.

Note

Outside the studied window, 2018–22 Cleveland (Baker Mayfield, David Njoku — both first-round) shows more offense and more volatility, echoing the Rams pattern.

Data & Variables

Sourced from nflfastR (play-by-play) and Spotrac (contracts). Volatility is modeled across three groups of predictors.

Continuity

Variable	Description
QBC	Same primary QB as prior season?
HCC	Same head coach as prior season?
OCC	Same offensive coordinator as prior season?

Team Characteristics

Variable	Description
Age	Average age of offensive starters
QBE	Primary QB's years of experience (rookie = 0)

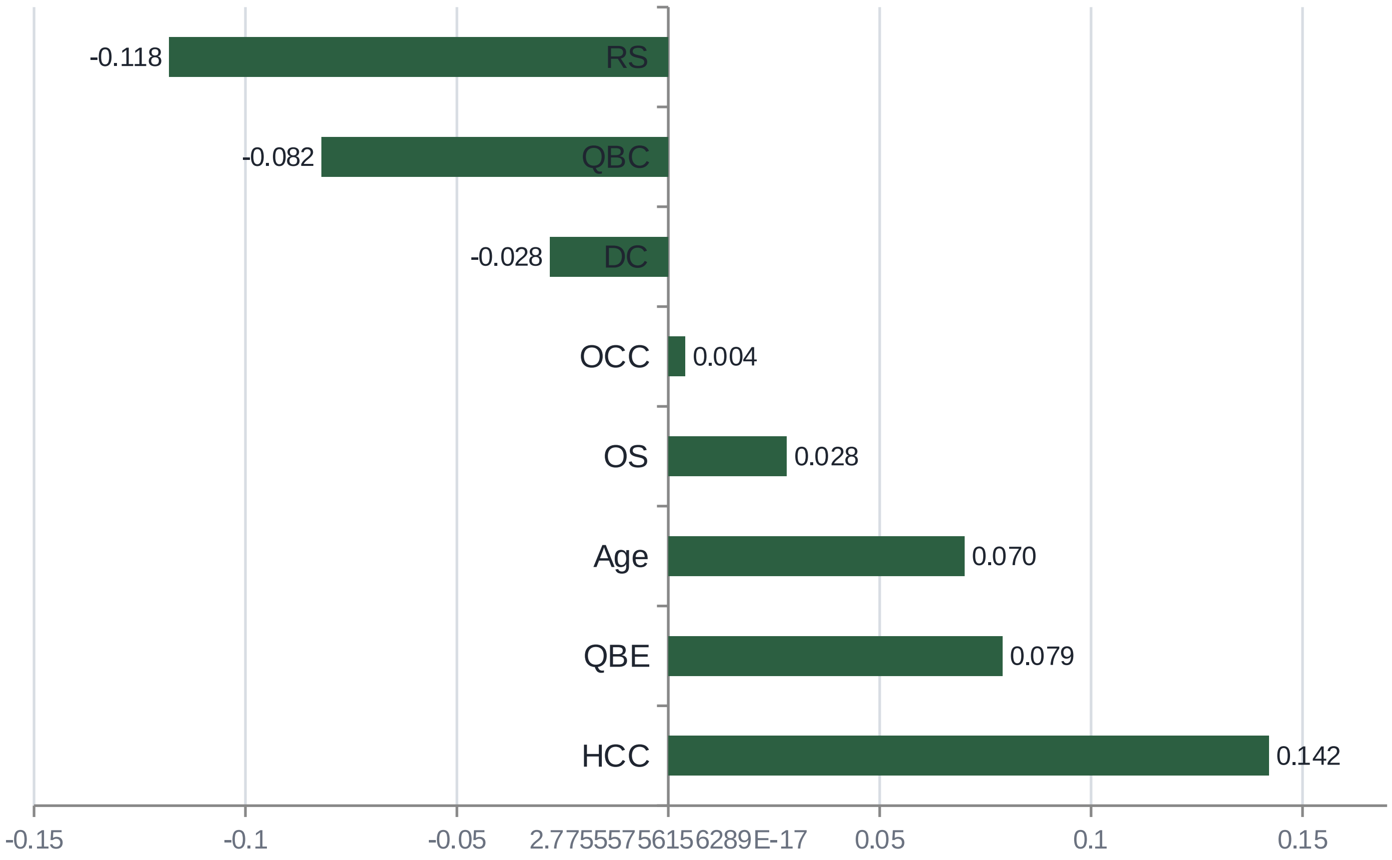
Front Office Decisions

Variable	Description
DC	Draft capital spent on offensive players (1/pick, summed)
RS	Share of cap space paid to rookies overall
OS	Share of cap space paid to all offensive players

Team and season fixed effects are included in every specification.

Exploratory Data Analysis

Raw correlation of each predictor with 4-year EPA/play volatility



All weak in isolation

Every bivariate correlation is small. That's expected — and it's exactly why the model adds squared and interaction terms.

- Coach continuity (HCC) shows the strongest raw pull toward higher volatility.
- Rookie cap share (RS) shows the strongest pull toward lower volatility.
- A weak straight-line correlation can still hide a real curved relationship.

Design Matrix

Predictors are standardized (_z) and entered with squared terms to let volatility rise, peak, and fall rather than move in one direction only. Two purpose-built interactions are added.

```
epa_volatility_4yr  ~  qbc + hcc + occ  +  off_dc_z (+sq)  +  age_z (+sq)  +  qb_exp_z (+sq)  +  rookie_prop_z (+sq)
+  offense_prop_z (+sq)  +  rookie_prop_z × offense_prop_z  +  rookie_prop_z × off_dc_z  +  team & season fixed
effects
```

Rookie Share × Draft Capital

A high rookie cap share means something different depending on who it's spent on — late-round depth vs. a premium first-round pick. This term tests whether concentration matters, not just the total.

Rookie Share × Offensive Share

A high rookie share paired with a high offensive cap commitment is a concentrated bet on youth. Paired with a low offensive share, it may just reflect a rebuild, not a strategy.

Two Estimators, One Specification

146 team-seasons · 47 predictors (df=98) · GVIF$^{1:2df1}$ ranges 1.29–1.86 across predictors — all under the 2.0 rule of thumb after standardization

OLS

Cluster-robust by team

Team & season fixed effects absorb stable franchise/era differences. Clustering standard errors by team guards against leftover within-team correlation across seasons.

Sensitive to a small number of high-leverage team-seasons.

Huber M-Estimation

Robust regression

Same fixed effects and predictors — but high-residual observations are automatically downweighted during fitting instead of dominating the coefficients.

Fit specifically to limit the influence of outlier team-seasons.

Where the two estimators agree on sign and significance, the finding is unlikely to be driven by a handful of extreme team-seasons. Where they disagree, the robust model's estimate is the more trustworthy read.

Results: OLS vs. Robust Regression

Selected coefficients — full model includes team & season fixed effects (R^2 = 0.686, Adj. R^2 = 0.535)

Variable	OLS	Robust (RLM)	Read
hcc (coach continuity)	+0.007 *	+0.006	*Significant, both directions agree*
qb_exp_z²	−0.006 ***	−0.006 **	*Significant curvature, confirmed*
offense_prop_z²	−0.003 **	−0.003 *	*Significant curvature, confirmed*
rookie_prop_z²	−0.002	−0.002 *	*RLM finds what OLS misses*
rookie_prop_z × off_dc_z	+0.004 *	+0.003	*OLS-only — treat as suggestive*
qbc (QB continuity)	−0.007	−0.004	*Not significant either way*

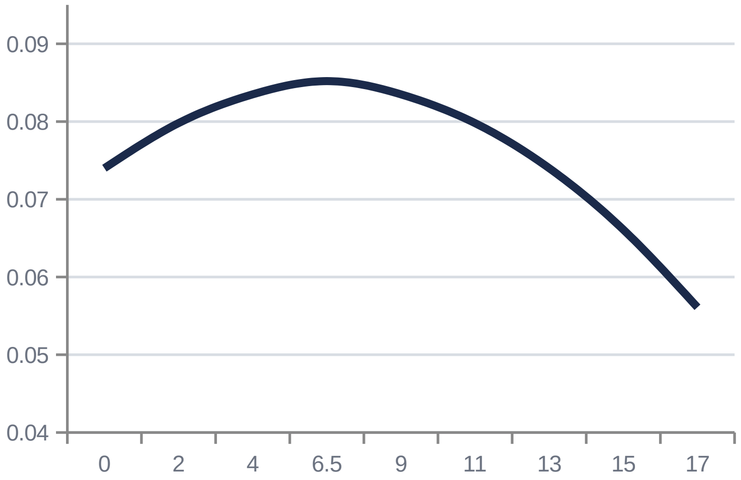
** p<0.1   ** p<0.05   *** p<0.01   Team fixed effects: significant. Season fixed effects: not significant.*

The Sweet Spots

Three predictors show a concave (rise-then-fall) relationship with volatility, not a straight line — curves computed directly from the fitted RLM coefficients

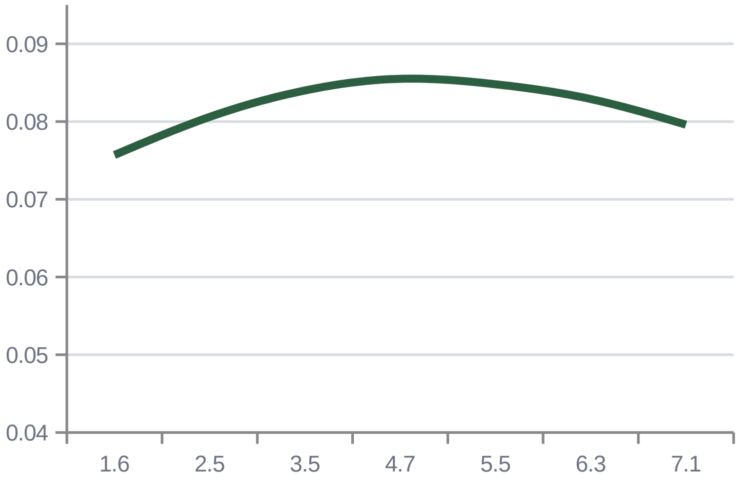
QB Experience

Peak volatility at 6.5 years



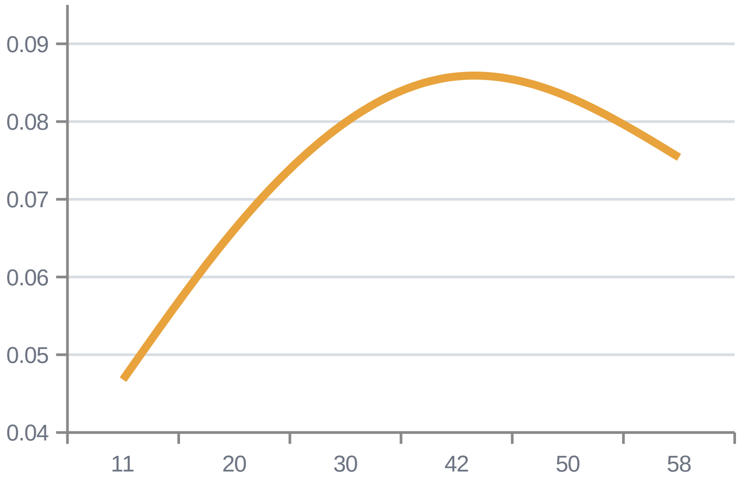
Rookie Cap Share

Peak volatility at 4.7%



Offensive Cap Share

Peak volatility at 42%



Reading the curves: extremes are stable, the middle is not.

Teams with a very green or very seasoned QB, a minimal or maximal rookie cap share, and a low or very high offensive cap commitment tend to be steadier. Volatility peaks in the middle — the zone teams often pass through mid-rebuild.

Diagnostics Hold Up

1

Residuals vs. Fitted

Scattered evenly above and below zero — no funnel shape, no sign of heteroskedasticity.

2

Cook's Distance

Flagged team-seasons cluster around real QB transitions (Denver 2011–13: Tebow → Manning), not random noise — the case for using the robust model.

3

Durbin-Watson: 2.039

No meaningful autocorrelation despite the repeated-team panel structure.

4

ACF of RLM Residuals

Every lag falls within confidence bounds except lag 7, which barely crosses — reasonable evidence against residual autocorrelation.

Key Takeaways

1. **Coach continuity matters most**

   Head-coach continuity shows the strongest, most consistent link to volatility of any variable measured — confirmed by both estimators.

2. **It's a curve, not a line**

   Rookie cap share, offensive cap share, and QB experience all show a rise-then-fall relationship with volatility — the middle of the range is the least stable place to be.

3. **Draft capital's role is suggestive, not settled**

   A big rookie investment concentrated in a premium pick shows a possible link to more volatility under OLS, but the robust model doesn't confirm it as strongly.

4. **Team identity still matters**

   Significant team fixed effects mean factors beyond roster spending — organizational culture, scheme, front-office philosophy — still shape offensive stability.